Stress Intensity at the Crack Tip

Lecture notes accompanying the interactive pages

Dr Régis Kubler
Arts et Métiers Institute of Technology

Written with the assistance of Claude Opus 5.5 (Anthropic)

October 2026

Abstract

These notes collect every equation and graph shown in the two interactive pages *Stress intensity at the crack tip* and *Angular functions* $f_{ij}(\theta)$. They follow the same order: stress concentration at an elliptical hole (Section 2), the crack-tip singularity (Section 3), the near-tip field and the definition of K (Section 4), handbook solutions (Section 5), extraction of K from numerical results (Section 6), the fracture criterion (Section 7), and the angular functions for all three modes with mixed-mode loading and the T-stress (Section 8). All figures were computed from the closed-form solutions given in the text by the script `make_figures.py` supplied with this document.

Contents

1 Coordinates and notation **2**

2 Stress concentration at an elliptical hole **2**
2.1 Geometry and the Inglis result . . . 2
2.2 Full field: Kolosov–Muskhelishvili solution . . . 3
2.3 From K_t to K: the sharp-crack limit . . . 3

3 The crack-tip singularity **5**
3.1 Westergaard solution for a centre crack . . . 5
3.2 Recovering the $1/\sqrt{r}$ term . . . 5

4 The near-tip field and the definition of K **6**

5 Calculating K from a handbook solution **6**
5.1 The general form . . . 6
5.2 Geometry factors used in the pages . . . 7
5.3 Worked example . . . 7

6 Calculating K from a numerical model **8**
6.1 Apparent K . . . 8
6.2 Why the apparent K drifts, and the extrapolation . . . 8
6.3 The J-integral alternative . . . 8

7 Using K: the fracture criterion **9**
7.1 Critical crack length . . . 9
7.2 The small-scale yielding principle . . . 9
7.3 Estimating the plastic zone . . . 9
7.4 When small-scale yielding holds . . . 10

8 Angular functions for all modes **11**
8.1 What depends on geometry and what does not 11
8.2 Mixed-mode loading 14
8.3 The T-stress 14

References **15**

1 Coordinates and notation

Throughout, the crack lies in the plane $y = 0$ and its front is parallel to z. Two coordinate systems are used (Figure 1).

- **Global coordinates** (x, y) for a through crack of length $2a$: the origin is the *centre* of the crack, so the crack occupies $-a \le x \le a$ and the tips are at $x = \pm a$.
- **Crack-tip polar coordinates** (r, θ): the origin is the *tip*; r is the distance from the tip and θ is measured anticlockwise from the crack line ahead of the tip. Hence $\theta = 0$ is straight ahead and $\theta = \pm\pi$ are the upper and lower crack faces. For the right-hand tip of a centre crack, $r = x - a$ on the line $y = 0$.

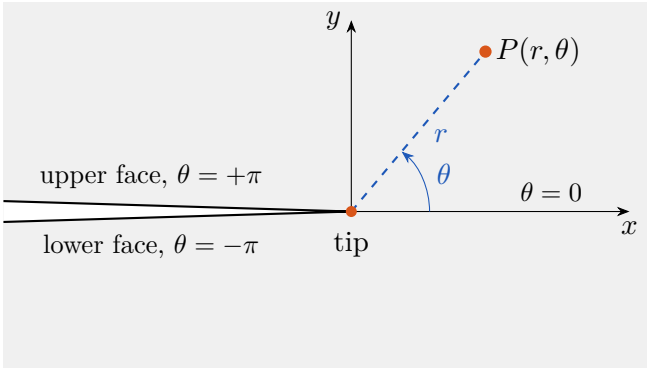


Figure 1: Crack-tip polar coordinates. The angular functions $f_{ij}(\theta)$ are always written in this frame, attached to the crack itself, whatever the orientation of the crack relative to the load.

Elastic constants: Young's modulus E, Poisson's ratio ν, shear modulus $\mu = E/2(1+\nu)$, and

$$E' = \begin{cases} E & \text{plane stress} \\ E/(1-\nu^2) & \text{plane strain} \end{cases}, \qquad \kappa = \begin{cases} (3-\nu)/(1+\nu) & \text{plane stress} \\ 3-4\nu & \text{plane strain.} \end{cases} \tag{1}$$

Units: stresses in MPa, lengths in metres inside $\sqrt{\pi a}$, so that K is in $\mathrm{MPa}\sqrt{\mathrm{m}}$.

2 Stress concentration at an elliptical hole

2.1 Geometry and the Inglis result

An infinite plate is loaded by a remote uniaxial stress σ_∞ in the y-direction and contains an elliptical hole with semi-axis a along x (across the load) and b along y. The radius of curvature at the end of the major axis is

$$\rho = \frac{b^2}{a}. \tag{2}$$

Inglis (1913) showed that the maximum stress, at $(x, y) = (\pm a, 0)$, is

$$\sigma_{\max} = \sigma_\infty \left(1 + \frac{2a}{b}\right) = \sigma_\infty \left(1 + 2\sqrt{\frac{a}{\rho}}\right), \qquad K_t = \frac{\sigma_{\max}}{\sigma_\infty} = 1 + 2\sqrt{\frac{a}{\rho}}. \tag{3}$$

A circular hole ($b = a$) gives the familiar $K_t = 3$; at the top of the hole $(0, \pm b)$ the hoop stress is $-\sigma_\infty$.

2.2 Full field: Kolosov–Muskhelishvili solution

The field plotted in Figure 2 is computed from complex potentials. Two complex variables are used:

- $z = x + iy = re^{i\theta}$ is the *physical* position in the plate, with the origin at the centre of the hole ($\bar{z} = x - iy$ is its conjugate);
- $\zeta = \rho_\zeta e^{i\eta}$ (zeta) is an auxiliary coordinate in a *mapped* plane, chosen so that the hole boundary becomes the unit circle $|\zeta| = 1$ and the plate becomes $|\zeta| \geq 1$.

Working in ζ turns the elliptical boundary into a circle, where the boundary condition is easy to impose. On the hole, $\zeta = e^{i\eta}$ and the mapping below gives $x = a\cos\eta$, $y = b\sin\eta$: the angle η is the eccentric angle of the ellipse, and $|\zeta| \to \infty$ corresponds to $|z| \to \infty$. The mapping is

$$z = \omega(\zeta) = R\left(\zeta + \frac{m}{\zeta}\right), \qquad R = \frac{a+b}{2}, \quad m = \frac{a-b}{a+b}, \tag{4}$$

with inverse $\zeta = \left(z + \sqrt{z^2 - 4R^2 m}\right)/2R$ (root with $|\zeta| \geq 1$). For uniaxial tension σ_∞ along y the potentials are

$$\varphi(\zeta) = \frac{\sigma_\infty R}{4}\left(\zeta - \frac{2+m}{\zeta}\right), \tag{5}$$

$$\psi(\zeta) = -\frac{\sigma_\infty R}{2}\left[-\zeta - \frac{1}{m\zeta} + \frac{(1+m)(1+m^2)}{m}\,\frac{\zeta}{\zeta^2 - m}\right], \tag{6}$$

and the stresses follow from the Kolosov–Muskhelishvili relations

$$\sigma_{xx} + \sigma_{yy} = 4\ \mathrm{Re}\ \Phi(z), \tag{7}$$

$$\sigma_{yy} - \sigma_{xx} + 2i\tau_{xy} = 2\left[\bar{z}\,\Phi'(z) + \Psi(z)\right], \tag{8}$$

with $\Phi = \varphi'(\zeta)/\omega'(\zeta)$, $\Phi'(z) = \Phi_{,\zeta}/\omega'(\zeta)$ and $\Psi = \psi'(\zeta)/\omega'(\zeta)$. Primes on φ, ψ, ω denote derivatives with respect to ζ; Φ and Ψ are the derivatives with respect to z, obtained by the chain rule $\mathrm{d}/\mathrm{d}z = (1/\omega')\,\mathrm{d}/\mathrm{d}\zeta$. Explicitly,

$$\omega'(\zeta) = R\left(1 - \frac{m}{\zeta^2}\right), \qquad \omega''(\zeta) = \frac{2Rm}{\zeta^3}, \tag{9}$$

$$\varphi'(\zeta) = \frac{\sigma_\infty R}{4}\left(1 + \frac{2+m}{\zeta^2}\right), \qquad \varphi''(\zeta) = -\frac{\sigma_\infty R}{2}\,\frac{2+m}{\zeta^3}, \tag{10}$$

$$\psi'(\zeta) = -\frac{\sigma_\infty R}{2}\left[-1 + \frac{1}{m\zeta^2} - \frac{(1+m)(1+m^2)}{m}\,\frac{\zeta^2 + m}{(\zeta^2 - m)^2}\right], \tag{11}$$

and

$$\Phi'(z) = \frac{1}{\omega'}\,\frac{\mathrm{d}\Phi}{\mathrm{d}\zeta} = \frac{\varphi''\omega' - \varphi'\omega''}{\omega'^3}. \tag{12}$$

These potentials were checked numerically: they give $\sigma_{yy}(a, 0) = \sigma_\infty(1 + 2a/b)$, $\sigma_{xx}(0, b) = -\sigma_\infty$, zero traction on the hole boundary, and $\sigma_{yy} \to \sigma_\infty$ far away.

2.3 From K_t to K: the sharp-crack limit

K_t depends only on the *shape* of the hole: scaling a and b together leaves $\sigma_{\max}$ unchanged. When $b \to 0$ the ellipse becomes a crack of length $2a$, $\rho \to 0$ and $K_t \to \infty$, so the stress concentration factor stops being useful. Multiplying (3) by $\frac{\sqrt{\pi}}{2}\sqrt{\rho}$,

$$\frac{\sqrt{\pi}}{2}\,\sigma_{\max}\sqrt{\rho} = \sigma_\infty\left(\sqrt{\pi a} + \frac{\sqrt{\pi\rho}}{2}\right), \tag{13}$$

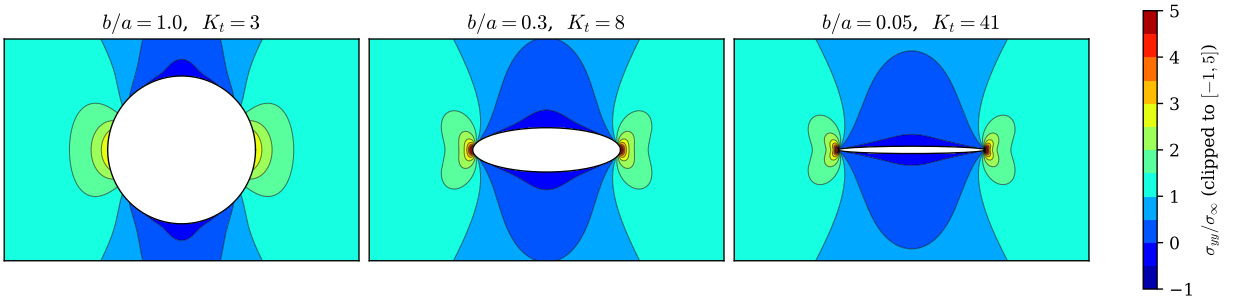


Figure 2: Opening stress $\sigma_{yy}/\sigma_\infty$ around an elliptical hole as it flattens from a circle into a crack. The peak at the end of the hole rises as $1 + 2a/b$.

which stays finite and, in the limit of a sharp crack, gives the stress intensity factor (Irwin):

$$\boxed{K_I = \lim_{\rho \to 0} \frac{\sqrt{\pi}}{2}\, \sigma_{\max} \sqrt{\rho} = \sigma_\infty \sqrt{\pi a}} \tag{14}$$

In normalised form the ratio plotted in Figure 3 is

$$\frac{(\sqrt{\pi}/2)\, \sigma_{\max} \sqrt{\rho}}{\sigma_\infty \sqrt{\pi a}} = 1 + \frac{1}{2}\sqrt{\frac{\rho}{a}} = 1 + \frac{b}{2a} \quad \xrightarrow{\rho \to 0} \quad 1. \tag{15}$$

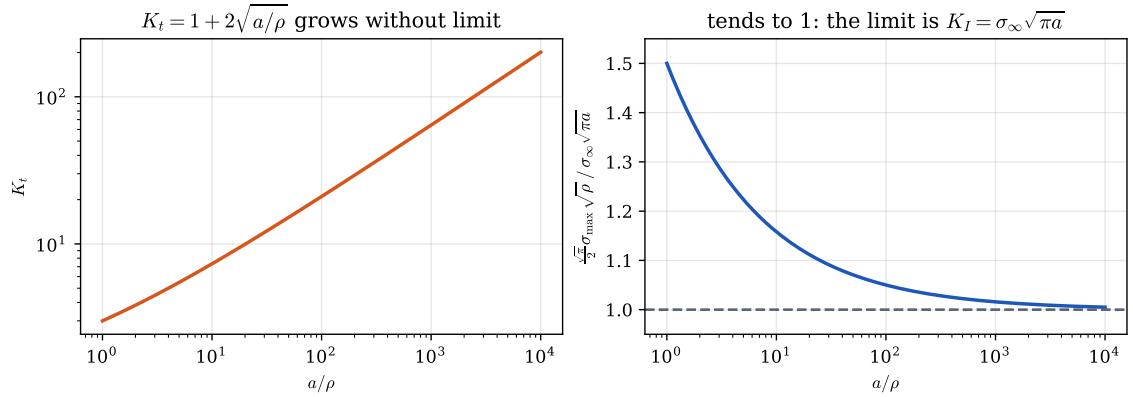


Figure 3: Left: K_t grows without limit as the hole sharpens. Right: the product $(\sqrt{\pi}/2)\sigma_{\max}\sqrt{\rho}$ tends to $\sigma_\infty\sqrt{\pi a}$, which is K_I for the crack.

Table 1: Stress concentration factor and stress intensity factor compared.

	K_t, stress concentration	K, stress intensity
Applies to	blunt notches and holes, ρ finite	sharp cracks, $\rho \to 0$
Units	none	$\mathrm{MPa}\sqrt{\mathrm{m}}$
Depends on	shape only	shape, size and load
Peak stress	finite, $\sigma_{\max} = K_t \sigma$	infinite, $\sigma \propto K/\sqrt{2\pi r}$
Compared with	yield or fatigue strength	fracture toughness K_{IC}

3 The crack-tip singularity

3.1 Westergaard solution for a centre crack

For a crack $-a \le x \le a$ in an infinite plate under remote equibiaxial stress σ_∞, Westergaard's stress function is

$$Z(z) = \frac{\sigma_\infty z}{\sqrt{z^2 - a^2}}, \qquad Z'(z) = -\frac{\sigma_\infty a^2}{(z^2 - a^2)^{3/2}}, \tag{16}$$

with the branch cut on the crack, and

$$\sigma_{xx} = \operatorname{Re} Z - y \operatorname{Im} Z', \qquad \sigma_{yy} = \operatorname{Re} Z + y \operatorname{Im} Z', \qquad \tau_{xy} = -y \operatorname{Re} Z'. \tag{17}$$

For uniaxial loading normal to the crack, subtract σ_∞ from σ_{xx}; σ_{yy} and τ_{xy} are unchanged. On the crack line ($y = 0$, $|x| > a$):

$$\sigma_{yy}(x, 0) = \frac{\sigma_\infty\, x}{\sqrt{x^2 - a^2}}, \tag{18}$$

where x is measured from the centre of the crack. This stress tends to infinity as $x \to a$.

3.2 Recovering the $1/\sqrt{r}$ term

Put $x = a + r$ in (18):

$$\sigma_{yy}(r, 0) = \frac{\sigma_\infty (a + r)}{\sqrt{r(2a + r)}} = \frac{\sigma_\infty \sqrt{\pi a}}{\sqrt{2\pi r}} \, \frac{1 + r/a}{\sqrt{1 + r/2a}} \xrightarrow{r \ll a} \frac{\sigma_\infty \sqrt{\pi a}}{\sqrt{2\pi r}} = \frac{K_I}{\sqrt{2\pi r}}. \tag{19}$$

Halving the distance from the tip raises the stress by $\sqrt{2}$, not by 2. The stress at the tip is therefore not a useful number for a sharp crack; the *amplitude* of the singularity, K_I, is.

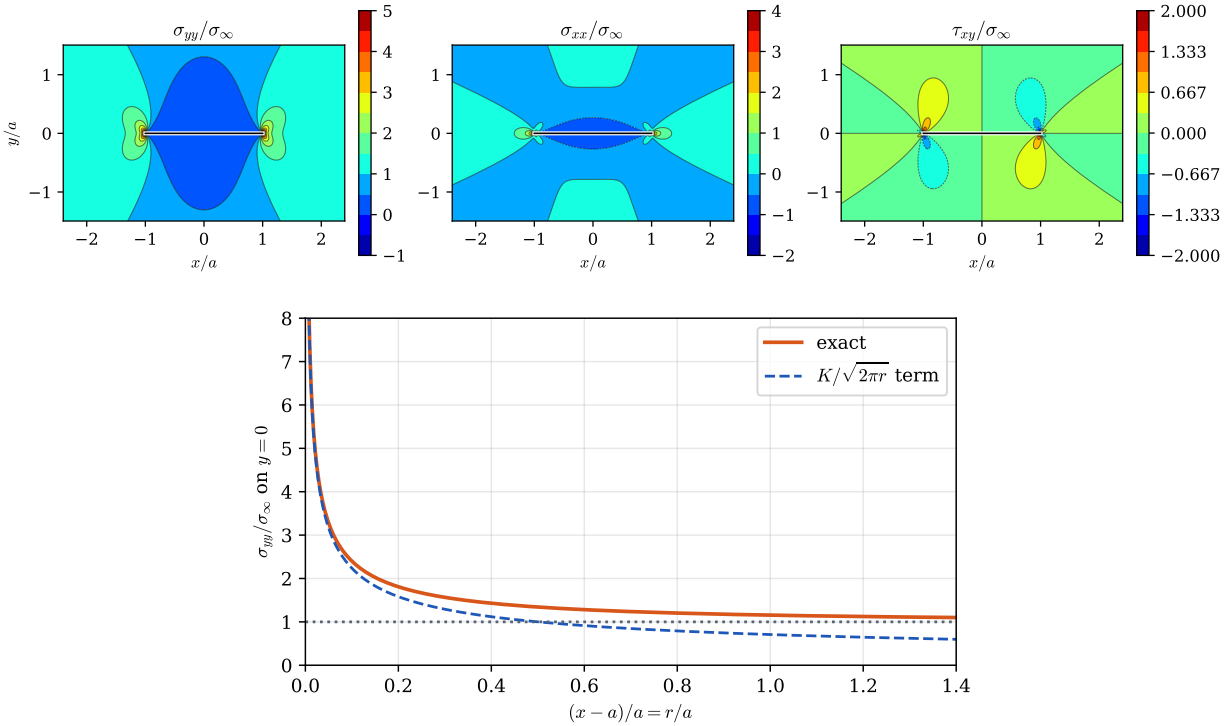


Figure 4: Top: Westergaard stress components around a centre crack under uniaxial tension (rainbow scale, blue = minimum, red = maximum, clipped near the tips). Bottom: σ_{yy} along the crack line ahead of the right tip, compared with its singular term $K_I/\sqrt{2\pi r}$.

4 The near-tip field and the definition of K

Williams (1957) showed that, close to the tip of any crack in a linear elastic body, the stresses take the form

$$\sigma_{ij}(r,\theta) = \frac{K}{\sqrt{2\pi r}}\, f_{ij}(\theta) + (\text{terms that are finite as } r \to 0). \tag{20}$$

For mode I,

$$f^I_{xx}(\theta) = \cos\tfrac{\theta}{2}\left[1 - \sin\tfrac{\theta}{2}\sin\tfrac{3\theta}{2}\right], \tag{21}$$

$$f^I_{yy}(\theta) = \cos\tfrac{\theta}{2}\left[1 + \sin\tfrac{\theta}{2}\sin\tfrac{3\theta}{2}\right], \tag{22}$$

$$f^I_{xy}(\theta) = \cos\tfrac{\theta}{2}\,\sin\tfrac{\theta}{2}\,\cos\tfrac{3\theta}{2}. \tag{23}$$

Geometry, crack length and load all enter through the single scalar K_I. Since $f^I_{yy}(0) = 1$,

$$K_I = \lim_{r\to 0}\ \sigma_{yy}(r,0)\sqrt{2\pi r}, \qquad \text{and in general} \qquad K_I = \frac{\sigma_{ij}(r,\theta)\sqrt{2\pi r}}{f_{ij}(\theta)} \quad \text{for } r \to 0,\ f_{ij} \neq 0. \tag{24}$$

The ratio is undefined where $f_{ij} = 0$, for example τ_{xy} on $\theta = 0$.

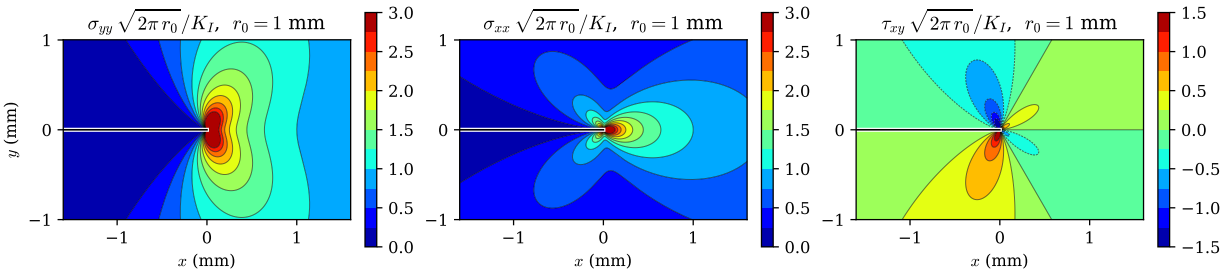


Figure 5: Mode I near-tip stress components $\sigma_{ij} = K_I f_{ij}(\theta)/\sqrt{2\pi r}$, normalised by $K_I/\sqrt{2\pi r_0}$ with $r_0 = 1\,\text{mm}$; the crack lies along the negative x-axis. Rainbow scale, blue = minimum, red = maximum.

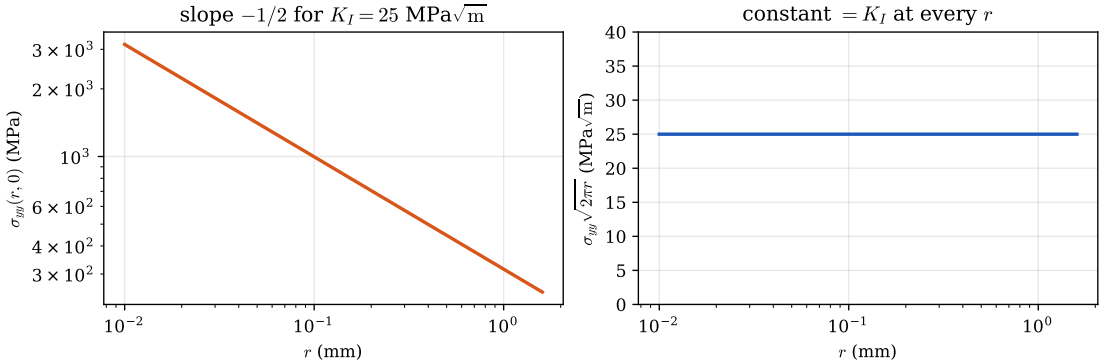


Figure 6: Along $\theta = 0$ the stress follows a slope of $-1/2$ on log–log axes, while $\sigma_{yy}\sqrt{2\pi r}$ stays equal to K_I at every distance. Example: $K_I = 25\,\text{MPa}\sqrt{\text{m}}$, r in mm.

5 Calculating K from a handbook solution

5.1 The general form

For standard geometries the elastic problem has been solved and tabulated as a dimensionless geometry factor Y (also written F or β):

$$K_I = Y\left(\frac{a}{W}\right)\sigma\sqrt{\pi a}. \tag{25}$$

Procedure:

1. identify the geometry and loading, and measure a as the handbook defines it (half-length for a centre crack, full depth for an edge crack);
2. read Y at the current a/W and check the range of validity;
3. convert a to metres;
4. multiply out $K_I = Y\sigma\sqrt{\pi a}$.

5.2 Geometry factors used in the pages

Table 2: Geometry factors for remote tension σ.

Geometry	Y	Validity
Through crack $2a$, infinite plate (Griffith)	1	exact
Edge crack a, semi-infinite plate	1.122	exact
Centre crack $2a$, width W (Feddersen)	$\sqrt{\sec(\pi a/W)}$	$2a/W \le 0.7$, $\pm 0.3\%$
Single edge crack a, width W (Tada)	$1.12 - 0.231\,\frac{a}{W} + 10.55\,(\frac{a}{W})^2 - 21.72\,(\frac{a}{W})^3 + 30.39\,(\frac{a}{W})^4$	$a/W \le 0.6$, $\pm 0.5\%$

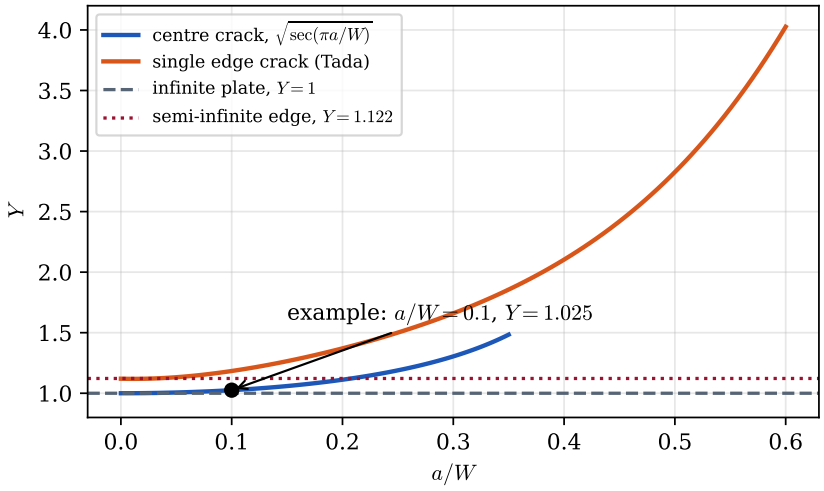


Figure 7: Geometry factors against a/W.

5.3 Worked example

Centre crack, $a = 20\,\text{mm}$, $W = 200\,\text{mm}$, $\sigma = 150\,\text{MPa}$:

$$\begin{aligned} a/W &= 20/200 = 0.100, \\ Y &= \sqrt{\sec(0.1\pi)} = 1.025, \\ K_I &= 1.025 \times 150\,\text{MPa} \times \sqrt{\pi \times 0.0200\,\text{m}} = 38.6\,\text{MPa}\sqrt{\text{m}}. \end{aligned}$$

6 Calculating K from a numerical model

6.1 Apparent K

When no handbook solution exists, the part is modelled (for example by finite elements) and K is extracted from the near-tip results. Each node gives an *apparent* K by inverting the singular term at its own distance r.

Stress method (nodes ahead of the tip, $\theta = 0$):

$$K_{\mathrm{app}}(r) = \sigma_{yy}(r, 0)\,\sqrt{2\pi r}. \tag{26}$$

Displacement method (nodes on the crack faces, $\theta = \pm\pi$). From the mode I displacement field (Table 5) the crack opening $\delta = u_y(r, \pi) - u_y(r, -\pi)$ is

$$\delta = \frac{K_I}{\mu}\sqrt{\frac{r}{2\pi}}\,(\kappa + 1) = \frac{8K_I}{E'}\sqrt{\frac{r}{2\pi}}, \qquad \text{so} \qquad K_{\mathrm{app}}(r) = \frac{E'\delta}{8}\sqrt{\frac{2\pi}{r}}. \tag{27}$$

6.2 Why the apparent K drifts, and the extrapolation

Higher-order terms make K_{app} vary with r. For the centre crack the exact Westergaard results are

$$\text{stress:} \quad K_{\mathrm{app}}(r) = \sigma\sqrt{\pi a}\,\frac{1 + r/a}{\sqrt{1 + r/2a}} \approx K_I\left(1 + \frac{3r}{4a}\right), \tag{28}$$

$$\text{displacement:} \quad K_{\mathrm{app}}(r) = \sigma\sqrt{\pi a}\,\sqrt{1 - \frac{r}{2a}} \approx K_I\left(1 - \frac{r}{4a}\right), \tag{29}$$

using the crack-face opening $\delta = (4\sigma/E')\sqrt{r(2a - r)}$. Both are nearly linear in r, so a straight line $K_{\mathrm{app}} = c_0 + c_1 r$ is fitted by least squares over the n points with $r \geq r_{\min}$,

$$c_1 = \frac{n\sum r_i K_i - \sum r_i \sum K_i}{n\sum r_i^2 - \left(\sum r_i\right)^2}, \qquad c_0 = \frac{\sum K_i - c_1 \sum r_i}{n}, \tag{30}$$

and the estimate is the intercept, $K_I \approx c_0$. Points nearest the tip are excluded because ordinary elements cannot reproduce the $1/\sqrt{r}$ variation.

Synthetic data used in the page and in Figure 8. $2a = 40\,\mathrm{mm}$, $\sigma = 120\,\mathrm{MPa}$, so $\sigma\sqrt{\pi a} = 30.08\,\mathrm{MPa}\sqrt{\mathrm{m}}$. Eighteen nodes at $r_i = 0.02\,(100)^{i/17}\,\mathrm{mm}$. A mesh error $\times\left(1 - 0.32\,e^{-r/0.07\,\mathrm{mm}}\right)$ (stress) or $\times\left(1 - 0.07\,e^{-r/0.07\,\mathrm{mm}}\right)$ (displacement) and $\pm 0.4\%$ scatter are added to the exact values.

Table 3: Extrapolated K from the synthetic data (reference $30.08\,\mathrm{MPa}\sqrt{\mathrm{m}}$).

Method	fit $r \geq 0.3\,\mathrm{mm}$	fit all points
Stress ahead of the tip	29.96 (-0.4%)	26.76 (-11.0%)
Crack opening behind the tip	30.01 (-0.2%)	29.36 (-2.4%)

6.3 The J-integral alternative

The J-integral is path-independent and is evaluated away from the tip, so it is less sensitive to the mesh. For linear elastic mode I loading $J = G = K_I^2/E'$, hence $K_I = \sqrt{JE'}$.

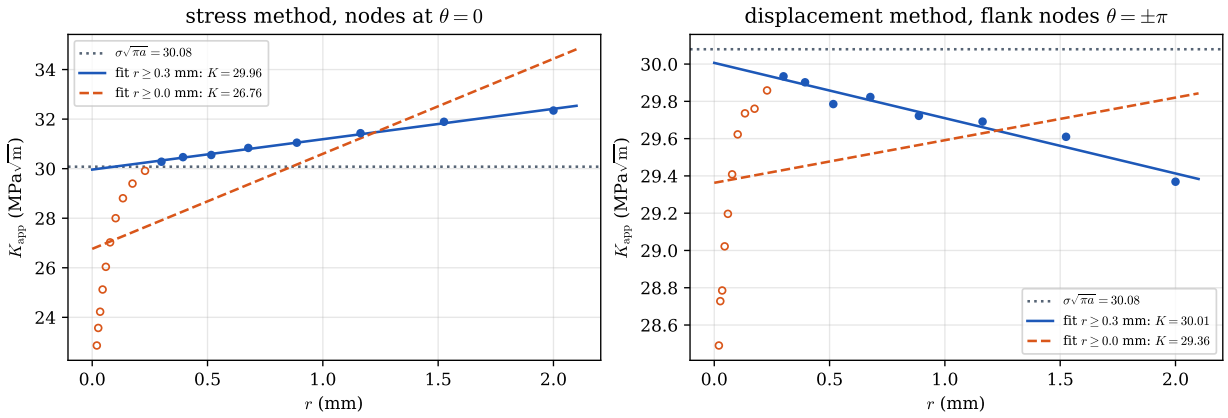


Figure 8: Extrapolation of the apparent K to $r = 0$. Open symbols are excluded from the solid fit; including them (dashed) biases the result.

7 Using K: the fracture criterion

7.1 Critical crack length

Because K fully describes the near-tip field, two cracks with the same K are in the same state. Fast fracture occurs when

$$K_I = Y\sigma\sqrt{\pi a} \;\geq\; K_{IC}, \qquad \Longrightarrow \qquad a_c = \frac{1}{\pi}\left(\frac{K_{IC}}{Y\sigma}\right)^2. \tag{31}$$

When Y depends on a/W this is solved numerically, for example by bisection on $Y(a/W)\sigma\sqrt{\pi a} - K_{IC} = 0$.

7.2 The small-scale yielding principle

The elastic solution predicts infinite stress at the tip, but a real metal yields. Linear elastic fracture mechanics remains useful because of the *small-scale yielding* (SSY) principle:

> If the plastic zone at the tip is small compared with the crack length, the remaining ligament and the thickness, it is embedded in a region where the elastic K field still holds. That surrounding K-dominated annulus then sets the boundary conditions on the plastic zone, so the state inside the plastic zone, and hence whether the crack grows, is controlled by K alone.

Two cracks in different components with the same K have identical plastic zones and fail at the same K, even though neither is perfectly elastic. This *autonomy* of the crack-tip region is what allows a toughness K_{IC} measured on a laboratory specimen to be used for a structure.

7.3 Estimating the plastic zone

First estimate. Setting the elastic opening stress on $\theta = 0$ equal to the yield strength,

$$\frac{K_I}{\sqrt{2\pi r_y}} = \sigma_{ys} \qquad \Longrightarrow \qquad r_y = \frac{1}{2\pi}\left(\frac{K_I}{\sigma_{ys}}\right)^2. \tag{32}$$

Irwin's correction. The elastic stress above σ_{ys} for $r < r_y$ (the shaded area in Figure 10) cannot be carried by the yielded material, so it is redistributed further out. Force balance makes the material behave as if the crack were longer, with an effective tip at $a_{\text{eff}} = a + r_y$, and doubles the zone:

$$r_p = 2r_y = \frac{1}{\pi}\left(\frac{K_I}{\sigma_{ys}}\right)^2 \quad \text{(plane stress)}, \qquad K_{\text{eff}} = Y(a_{\text{eff}})\,\sigma\sqrt{\pi a_{\text{eff}}}. \tag{33}$$

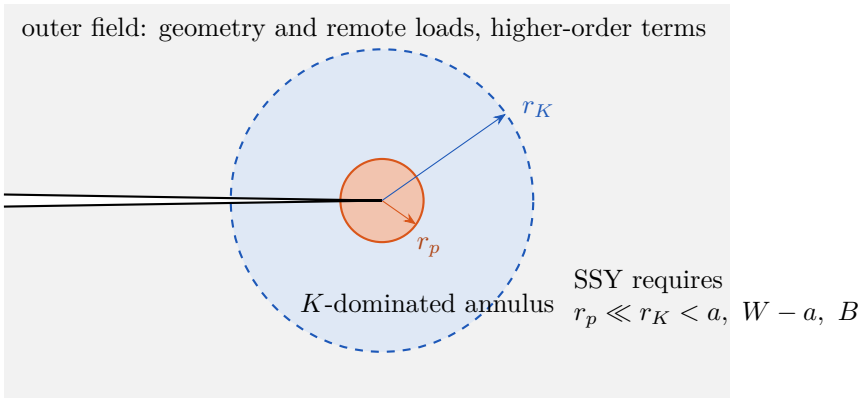


Figure 9: Small-scale yielding: the plastic zone (radius $\sim r_p$) sits well inside the annulus where $\sigma_{ij} \approx K f_{ij}(\theta)/\sqrt{2\pi r}$ is accurate. Outside r_K the higher-order terms of the Williams series matter.

Plane strain. Triaxial constraint raises the stress needed to yield, roughly to $\sqrt{3}\,\sigma_{ys}$ ahead of the tip, giving

$$r_p \approx \frac{1}{3\pi}\left(\frac{K_I}{\sigma_{ys}}\right)^2 \qquad \text{(plane strain)}. \tag{34}$$

Strip-yield (Dugdale) model. Treating the plastic zone as a thin strip closed by a cohesive stress σ_{ys} gives, for small σ/σ_{ys},

$$\rho_D = \frac{\pi}{8}\left(\frac{K_I}{\sigma_{ys}}\right)^2, \tag{35}$$

close to Irwin's r_p ($\pi/8 = 0.39$ against $1/\pi = 0.32$).

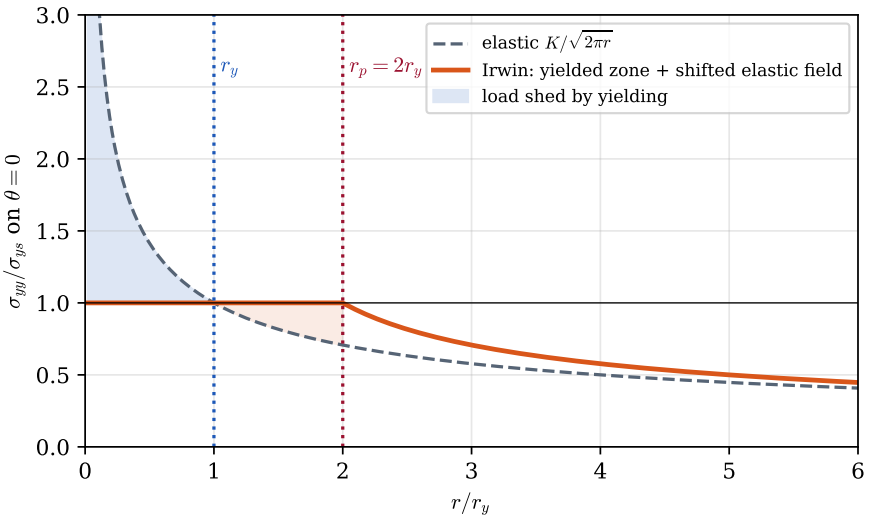


Figure 10: Opening stress ahead of the tip with Irwin's correction. Yielding caps the stress at σ_{ys}; the load shed for $r < r_y$ is carried between r_y and $r_p = 2r_y$, and beyond r_p the elastic field is that of a crack with its tip moved forward by r_y.

7.4 When small-scale yielding holds

- The plastic zone must be small compared with every in-plane length: $r_p \ll a$ and $r_p \ll W - a$. A common rule of thumb is $r_p \lesssim a/50$ for an accurate K, and a plastic zone above about $0.1\,a$ makes LEFM doubtful. The pages flag this second limit.

- The applied stress must stay well below yield, typically $\sigma \lesssim 0.8\,\sigma_{ys}$ on the net section, otherwise the whole ligament yields.
- For a valid plane-strain K_{IC} (ASTM E399) every dimension must be about 25 times the plane-strain plastic zone:

$$a,\ B,\ (W-a)\ \geq\ 2.5\left(\frac{K_{IC}}{\sigma_{ys}}\right)^2. \tag{36}$$

When these conditions fail the regime is *large-scale yielding*: K no longer controls the crack tip, and elastic–plastic parameters such as the J-integral or the crack-tip opening displacement (CTOD) must be used instead. In the small-scale yielding limit they are related to K by $J = K_I^2/E'$ and $\delta_t \approx K_I^2/(m\,\sigma_{ys}E')$ with $m \approx 1$ (plane stress) to 2 (plane strain).

Table 4: Typical teaching values (use certified data for design) and critical half crack length for the centre crack with $W = 200\,\text{mm}$, $\sigma = 150\,\text{MPa}$.

Material	K_{IC} ($\text{MPa}\sqrt{\text{m}}$)	σ_{ys} (MPa)	a_c (mm)
Al 7075-T6	29	505	11.7
AISI 4340, tempered 260 °C	50	1500	31.2
Ti-6Al-4V, annealed	66	880	46.1
PMMA	1.0	70	($\sigma > \sigma_{ys}$: LEFM not applicable)

Example (Ti-6Al-4V). At $a = 20\,\text{mm}$, $K_I = 38.6\,\text{MPa}\sqrt{\text{m}}$, so $K_{IC}/K_I = 1.71$ and $r_p = (1/\pi)(38.6/880)^2 = 0.61\,\text{mm}$, which is 3% of a: small-scale yielding holds. A valid K_{IC} test needs $B \geq 2.5(66/880)^2 = 14.1\,\text{mm}$.

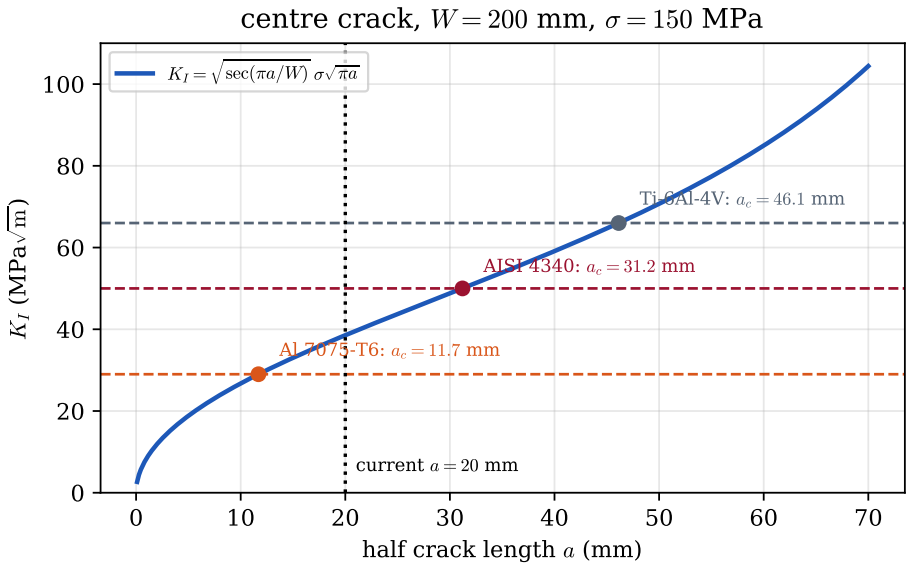


Figure 11: K_I against crack length compared with K_{IC} for three materials.

8 Angular functions for all modes

8.1 What depends on geometry and what does not

The angular functions $f_{ij}(\theta)$ are the same for every geometry; they change only with the *mode* of loading:

- **Mode I** (opening): tension normal to the crack plane;

- **Mode II** (in-plane sliding): shear parallel to the crack, normal to the front;
- **Mode III** (tearing): out-of-plane shear, parallel to the front.

Geometry and load enter only through K_I, K_{II}, K_{III} and through the higher-order terms, starting with the T-stress. This remains true for an inclined crack, provided θ is measured in the crack's own frame (Figure 1). It does not hold for V-notches (the exponent and angular functions depend on the notch angle), interface cracks between dissimilar materials, or anisotropic materials.

Table 5: Mode I near-tip field.

σ_{xx}	$\dfrac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right]$
σ_{yy}	$\dfrac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right]$
τ_{xy}	$\dfrac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\sin\frac{\theta}{2}\cos\frac{3\theta}{2}$
σ_{rr}	$\dfrac{K_I}{\sqrt{2\pi r}}\left[\frac{5}{4}\cos\frac{\theta}{2} - \frac{1}{4}\cos\frac{3\theta}{2}\right]$
$\sigma_{\theta\theta}$	$\dfrac{K_I}{\sqrt{2\pi r}}\left[\frac{3}{4}\cos\frac{\theta}{2} + \frac{1}{4}\cos\frac{3\theta}{2}\right]$
$\tau_{r\theta}$	$\dfrac{K_I}{\sqrt{2\pi r}}\left[\frac{1}{4}\sin\frac{\theta}{2} + \frac{1}{4}\sin\frac{3\theta}{2}\right]$
σ_{zz}	0 (plane stress), $\nu(\sigma_{xx} + \sigma_{yy})$ (plane strain)
u_x	$\dfrac{K_I}{2\mu}\sqrt{\dfrac{r}{2\pi}} \cos\frac{\theta}{2}\left[\kappa - 1 + 2\sin^2\frac{\theta}{2}\right]$
u_y	$\dfrac{K_I}{2\mu}\sqrt{\dfrac{r}{2\pi}} \sin\frac{\theta}{2}\left[\kappa + 1 - 2\cos^2\frac{\theta}{2}\right]$

Table 6: Mode II near-tip field.

σ_{xx}	$-\dfrac{K_{II}}{\sqrt{2\pi r}} \sin\frac{\theta}{2}\left[2 + \cos\frac{\theta}{2}\cos\frac{3\theta}{2}\right]$
σ_{yy}	$\dfrac{K_{II}}{\sqrt{2\pi r}} \sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{3\theta}{2}$
τ_{xy}	$\dfrac{K_{II}}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right]$
σ_{rr}	$\dfrac{K_{II}}{\sqrt{2\pi r}}\left[-\frac{5}{4}\sin\frac{\theta}{2} + \frac{3}{4}\sin\frac{3\theta}{2}\right]$
$\sigma_{\theta\theta}$	$\dfrac{K_{II}}{\sqrt{2\pi r}}\left[-\frac{3}{4}\sin\frac{\theta}{2} - \frac{3}{4}\sin\frac{3\theta}{2}\right]$
$\tau_{r\theta}$	$\dfrac{K_{II}}{\sqrt{2\pi r}}\left[\frac{1}{4}\cos\frac{\theta}{2} + \frac{3}{4}\cos\frac{3\theta}{2}\right]$
σ_{zz}	0 (plane stress), $\nu(\sigma_{xx} + \sigma_{yy})$ (plane strain)
u_x	$\dfrac{K_{II}}{2\mu}\sqrt{\dfrac{r}{2\pi}} \sin\frac{\theta}{2}\left[\kappa + 1 + 2\cos^2\frac{\theta}{2}\right]$
u_y	$-\dfrac{K_{II}}{2\mu}\sqrt{\dfrac{r}{2\pi}} \cos\frac{\theta}{2}\left[\kappa - 1 - 2\sin^2\frac{\theta}{2}\right]$

The polar components follow from the Cartesian ones by rotation, $\sigma_{rr} = \sigma_{xx}\cos^2\theta + \sigma_{yy}\sin^2\theta + 2\tau_{xy}\sin\theta\cos\theta$, $\sigma_{\theta\theta} = \sigma_{xx}\sin^2\theta + \sigma_{yy}\cos^2\theta - 2\tau_{xy}\sin\theta\cos\theta$, $\tau_{r\theta} = (\sigma_{yy} - \sigma_{xx})\sin\theta\cos\theta + \tau_{xy}(\cos^2\theta - \sin^2\theta)$; the tables above were checked against this rotation numerically.

Table 7: Mode III near-tip field (all in-plane stresses vanish).

τ_{xz}	$-\dfrac{K_{III}}{\sqrt{2\pi r}}\sin\dfrac{\theta}{2}$
τ_{yz}	$\dfrac{K_{III}}{\sqrt{2\pi r}}\cos\dfrac{\theta}{2}$
τ_{rz}	$\dfrac{K_{III}}{\sqrt{2\pi r}}\sin\dfrac{\theta}{2}$
$\tau_{\theta z}$	$\dfrac{K_{III}}{\sqrt{2\pi r}}\cos\dfrac{\theta}{2}$
u_z	$\dfrac{2K_{III}}{\mu}\sqrt{\dfrac{r}{2\pi}}\sin\dfrac{\theta}{2}$

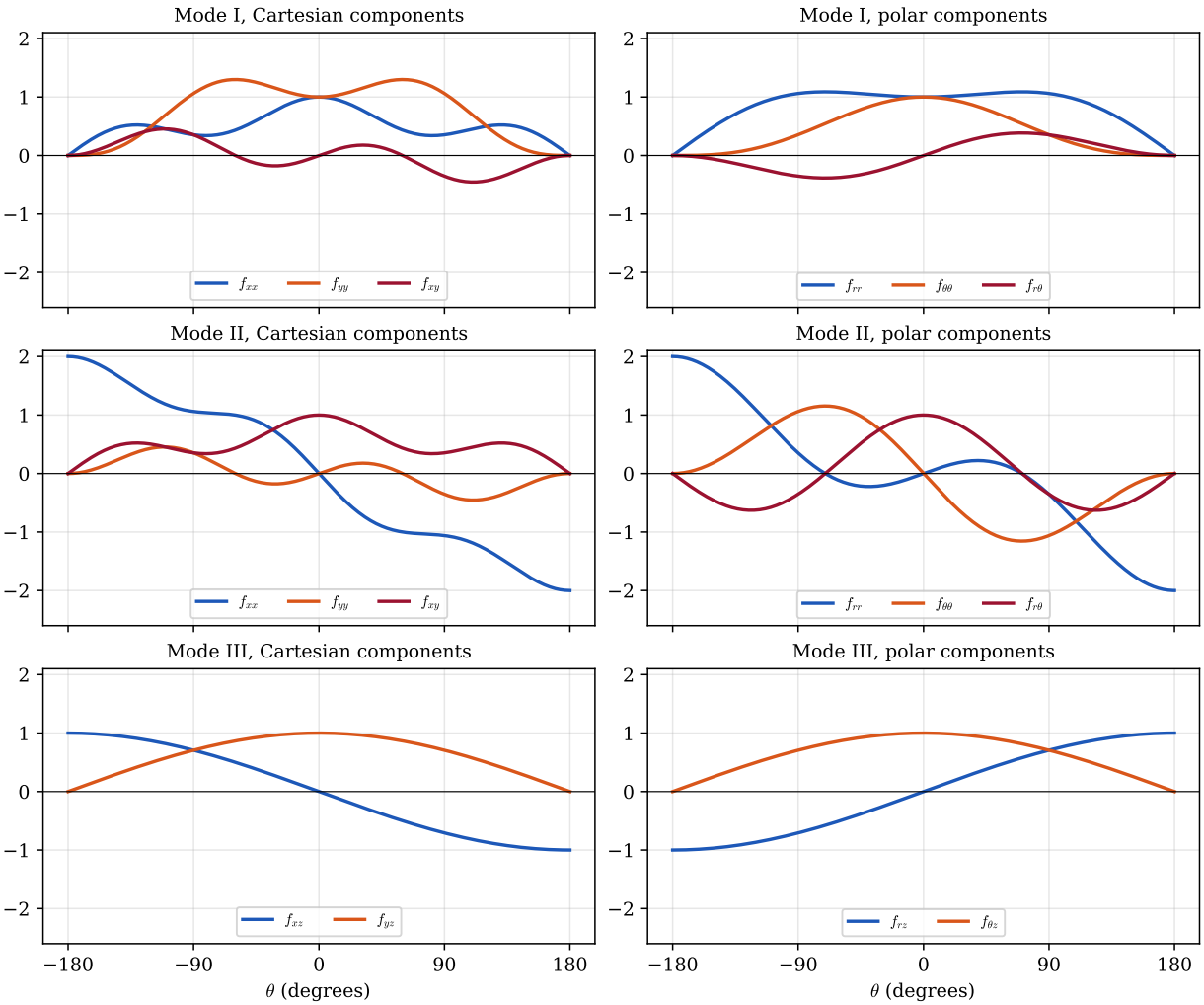


Figure 12: Angular functions $f_{ij}(\theta)$ for the three modes, from the lower crack face ($\theta = -180°$) to the upper crack face ($\theta = +180°$).

8.2 Mixed-mode loading

In linear elasticity the three fields superpose, each with its own K:

$$\sigma_{ij} = \frac{1}{\sqrt{2\pi r}} \left[K_I f_{ij}^{I}(\theta) + K_{II} f_{ij}^{II}(\theta) + K_{III} f_{ij}^{III}(\theta) \right]. \tag{37}$$

The K's themselves cannot be added because the angular functions differ, but the energy release rate does add:

$$G = \frac{K_I^2 + K_{II}^2}{E'} + \frac{K_{III}^2}{2\mu}. \tag{38}$$

Table 8: Loading cases and the modes they produce (through crack $2a$, infinite plate unless stated).

Geometry and loading	Mode	Stress intensity factors
Tension normal to the crack plane	I	$K_I = Y\sigma\sqrt{\pi a}$
In-plane shear τ parallel to the crack	II	$K_{II} = \tau\sqrt{\pi a}$
Anti-plane shear τ (torsion of a cracked shaft)	III	$K_{III} = \tau\sqrt{\pi a}$
Crack at angle β to uniaxial σ	I + II	$K_I = \sigma\sqrt{\pi a}\sin^2\beta$, $K_{II} = \sigma\sqrt{\pi a}\sin\beta\cos\beta$
Tension plus torsion on a cracked tube	I + III	superpose the two solutions

For the inclined crack, β is the angle between the crack line and the load: $\beta = 90^\circ$ gives pure mode I and $\beta = 45^\circ$ gives $K_I = K_{II} = \frac{1}{2}\sigma\sqrt{\pi a}$.

8.3 The T-stress

The full Williams expansion for mode I is

$$\sigma_{ij} = \frac{K_I}{\sqrt{2\pi r}} f_{ij}^{I}(\theta) + T\,\delta_{1i}\delta_{1j} + O(\sqrt{r}). \tag{39}$$

T is a constant stress parallel to the crack (it adds only to σ_{xx}). It depends on geometry and loading, and it controls crack-tip constraint and the size of the zone in which K alone describes the field.

Table 9: T-stress for a through crack $2a$ in an infinite plate.

Loading	T
Uniaxial σ normal to the crack	$-\sigma$
Equibiaxial σ	0
Uniaxial σ at angle β to the crack line	$\sigma\cos 2\beta$

For finite specimens T is usually given through the biaxiality ratio $B = T\sqrt{\pi a}/K_I$ (Leevers and Radon 1982; Sherry et al. 1995).

Physical meaning. T is a uniform stress acting parallel to the crack. It is the same at every distance r and is set by the remote loading and the geometry, not by the singularity. Deep edge cracks in bending (SENB, CT) have $T > 0$; short cracks and centre-cracked panels in tension have $T < 0$.

Why T does not change J or G. A stress parallel to the crack does not open the crack faces, so it releases no energy at the tip. In the J-integral the T^2 part is a uniform field with no singularity inside the contour and integrates to zero, and the $K_I T$ cross term cancels between the upper and lower halves of the contour by symmetry. Hence $J = K_I^2/E'$ for any T; only the split between the $\int W \,\mathrm{d}y$ and $-\int T_i \,\partial u_i/\partial x \,\mathrm{d}s$ terms changes. (Here $T_i = \sigma_{ij} n_j$ is the traction vector, a different quantity from the scalar T-stress.)

Constraint. T changes the stress triaxiality ahead of the tip (Larsson and Carlsson 1973).

- $T > 0$, high constraint: higher hydrostatic stress, a smaller plastic zone that leans forward, and toughness close to the lower bound. Deep-cracked test specimens are in this regime.
- $T < 0$, loss of constraint: lower hydrostatic stress, a larger plastic zone, and an apparent toughness above K_{IC} or J_{Ic}. Shallow cracks and thin panels in tension are in this regime.

Two cracks with the same K or J can therefore fail at different loads. This is the basis of two-parameter fracture mechanics: K–T or J–T (Betegón and Hancock 1991) and J–Q (O'Dowd and Shih 1991).

Crack path. T also controls directional stability: with $T > 0$ a straight crack tends to kink out of its plane, with $T < 0$ it stays straight (Cotterell and Rice 1980).

References

- C. E. Inglis (1913). Stresses in a plate due to the presence of cracks and sharp corners. *Trans. Inst. Naval Architects* 55, 219–241.
- N. I. Muskhelishvili (1953). *Some Basic Problems of the Mathematical Theory of Elasticity.* Noordhoff.
- H. M. Westergaard (1939). Bearing pressures and cracks. *J. Appl. Mech.* 6, A49–A53.
- M. L. Williams (1957). On the stress distribution at the base of a stationary crack. *J. Appl. Mech.* 24, 109–114.
- G. R. Irwin (1957). Analysis of stresses and strains near the end of a crack traversing a plate. *J. Appl. Mech.* 24, 361–364.
- H. Tada, P. C. Paris, G. R. Irwin (2000). *The Stress Analysis of Cracks Handbook*, 3rd ed. ASME.
- T. L. Anderson (2017). *Fracture Mechanics: Fundamentals and Applications*, 4th ed. CRC Press.
- ASTM E399. Standard test method for linear-elastic plane-strain fracture toughness of metallic materials.
- C. E. Feddersen (1967). Discussion of plane strain crack toughness testing. *ASTM STP 410*, 77–79.
- D. S. Dugdale (1960). Yielding of steel sheets containing slits. *J. Mech. Phys. Solids* 8, 100–104.
- P. S. Leevers, J. C. Radon (1982). Inherent stress biaxiality in various fracture specimen geometries. *Int. J. Fract.* 19, 311–325.
- A. H. Sherry, C. C. France, M. R. Goldthorpe (1995). Compendium of T-stress solutions for two and three dimensional cracked geometries. *Fatigue Fract. Eng. Mater. Struct.* 18, 141–155.
- J. R. Rice (1968). A path independent integral and the approximate analysis of strain concentration by notches and cracks. *J. Appl. Mech.* 35, 379–386.
- S. G. Larsson, A. J. Carlsson (1973). Influence of non-singular stress terms and specimen geometry on small-scale yielding at crack tips in elastic–plastic materials. *J. Mech. Phys. Solids* 21, 263–277.
- C. Betegón, J. W. Hancock (1991). Two-parameter characterization of elastic–plastic crack-tip fields. *J. Appl. Mech.* 58, 104–110.
- N. P. O'Dowd, C. F. Shih (1991). Family of crack-tip fields characterized by a triaxiality parameter—I. Structure of fields. *J. Mech. Phys. Solids* 39, 989–1015.

- B. Cotterell, J. R. Rice (1980). Slightly curved or kinked cracks. *Int. J. Fract.* 16, 155–169.
- A. A. Griffith (1921). The phenomena of rupture and flow in solids. *Phil. Trans. R. Soc. A* 221, 163–198.